MIS 600 · DATABASE MANAGEMENT SYSTEMS

Chapter 3

The Enhanced E-R Model

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Fall 2026

Text: Modern Database Management, 14th ed. (Hoffer, Topi, Venkataraman, and Bala, 2026). Figures in this deck come from Chapter 3 and are used for instruction in this course.

The plan for tonight

- First half. Supertypes and subtypes: the problem they solve, the notation, and when to use them.
- Finding them. Generalization and specialization.
- The rules. Completeness, disjointness, and discriminators.
- Break.
- Second half. Hierarchies, then the Pine Valley Furniture diagram rebuilt.
- The big picture. Entity clustering, packaged data models, and the data models behind analytics.

Have Draw.io open when we reach Pine Valley.

What you should be able to do after this chapter

- Recognize when a supertype/subtype relationship belongs in a model
- Use specialization and generalization to define one
- Specify completeness and disjointness constraints, and add a subtype discriminator
- Develop a supertype/subtype hierarchy for a realistic business situation
- Develop an entity cluster to simplify a large E-R diagram
- Explain the structures of a universal (packaged) data model, and how a project that starts from one differs
- Explain the data modeling structures used for analytical applications

PART ONE

Supertypes and subtypes

Why the E-R model needed an upgrade

- The basic E-R model dates from the mid-1970s. It still handles most business problems well.
- Business got more complex: segmented markets, customized products, decisions driven by analytics
- The enhanced E-R (EER) model adds constructs for that complexity. The big one is the supertype/subtype relationship.
- EER sits close to object-oriented modeling (Elmasri and Navathe, 2021, as cited in the text)
- Tonight also covers entity clustering, packaged data models, and a preview of the models used for analytics

"Redundancy = bad. Simplicity = good."

Entities that are almost the same

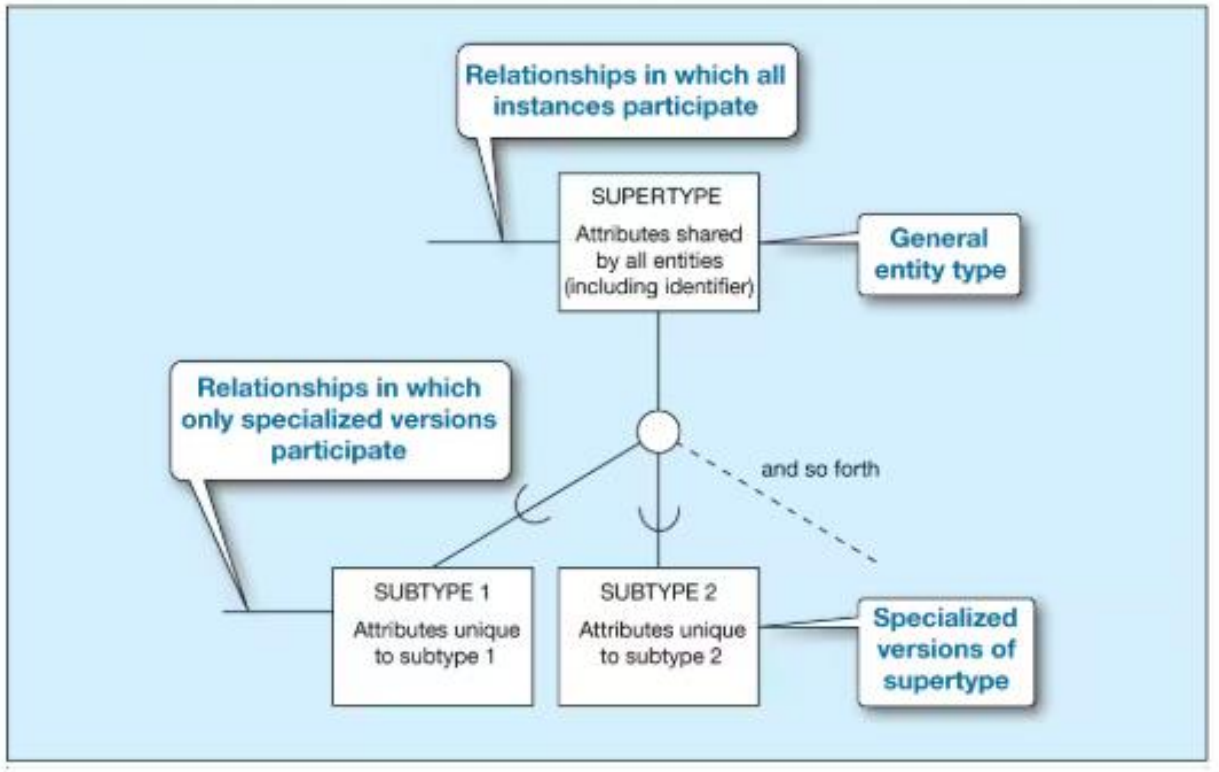
- An entity type is a set of instances that share properties. Chapter 2 already allowed optional attributes.
- The hard case: instances that share most properties, but each kind has something of its own the business cares about
- Subtype: a subgrouping of the entities in an entity type that is meaningful to the organization
- Supertype: a generic entity type that has a relationship with one or more subtypes
- STUDENT with GRADUATE STUDENT and UNDERGRADUATE STUDENT. CAR with SEDAN, SPORTS CAR, and COUPE.
- This room is the example. Same roster, same exams, and graduate students add an extension. STUDENT is the supertype.

They were hiding in Chapter 2. Why can a Pine Valley customer do business in no territory? Two kinds of customers, and only one kind is assigned. The old diagram could not say so.

The notation

Figure 3-1(a). The supertype connects to a circle, and the circle connects to each subtype.

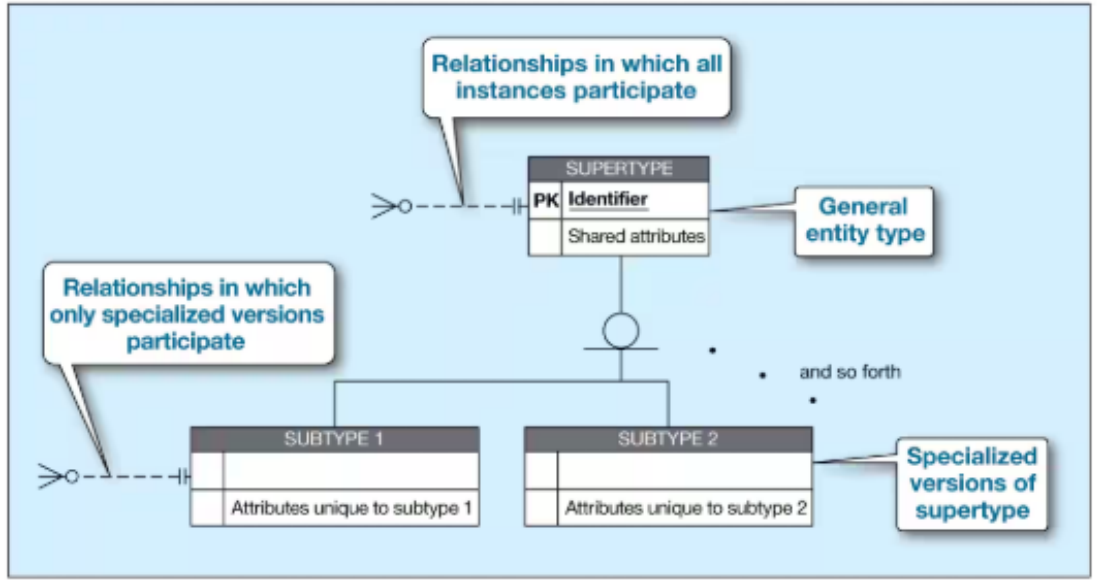
- Shared attributes, including the identifier, sit in the supertype
- Attributes unique to one subtype sit in that subtype. Same rule for relationships.
- The small U says subset and points the direction. It is optional, and the book usually leaves it out.
- Relationships that all instances join attach to the supertype. Relationships that only some join attach to the subtype.
- The circle is not a relationship. No verb phrase, no crow's feet.



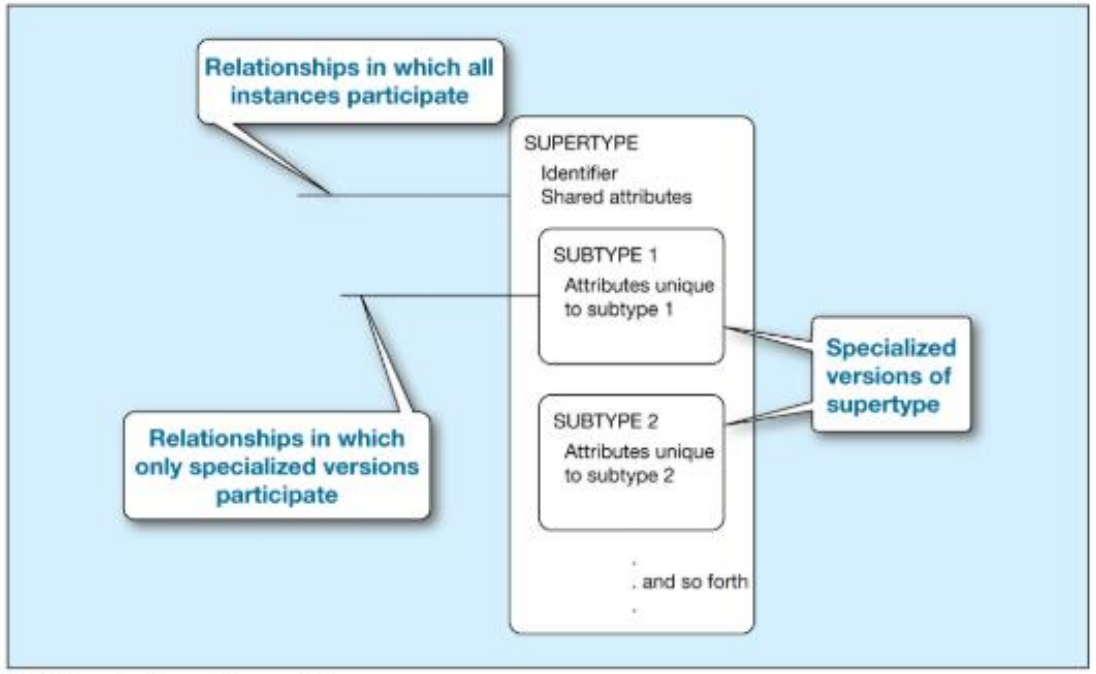
(a) EER notation

Two other notations you will meet

Figure 3-1(b) and (c). Same information, drawn differently.



(b) Microsoft Visio notation



(c) Subtypes inside supertypes notation

(b) Microsoft Visio: a PK row and a category circle. The Pine Valley figures use this.

(c) Subtypes drawn inside the supertype. CASE tools and published universal data models use this.

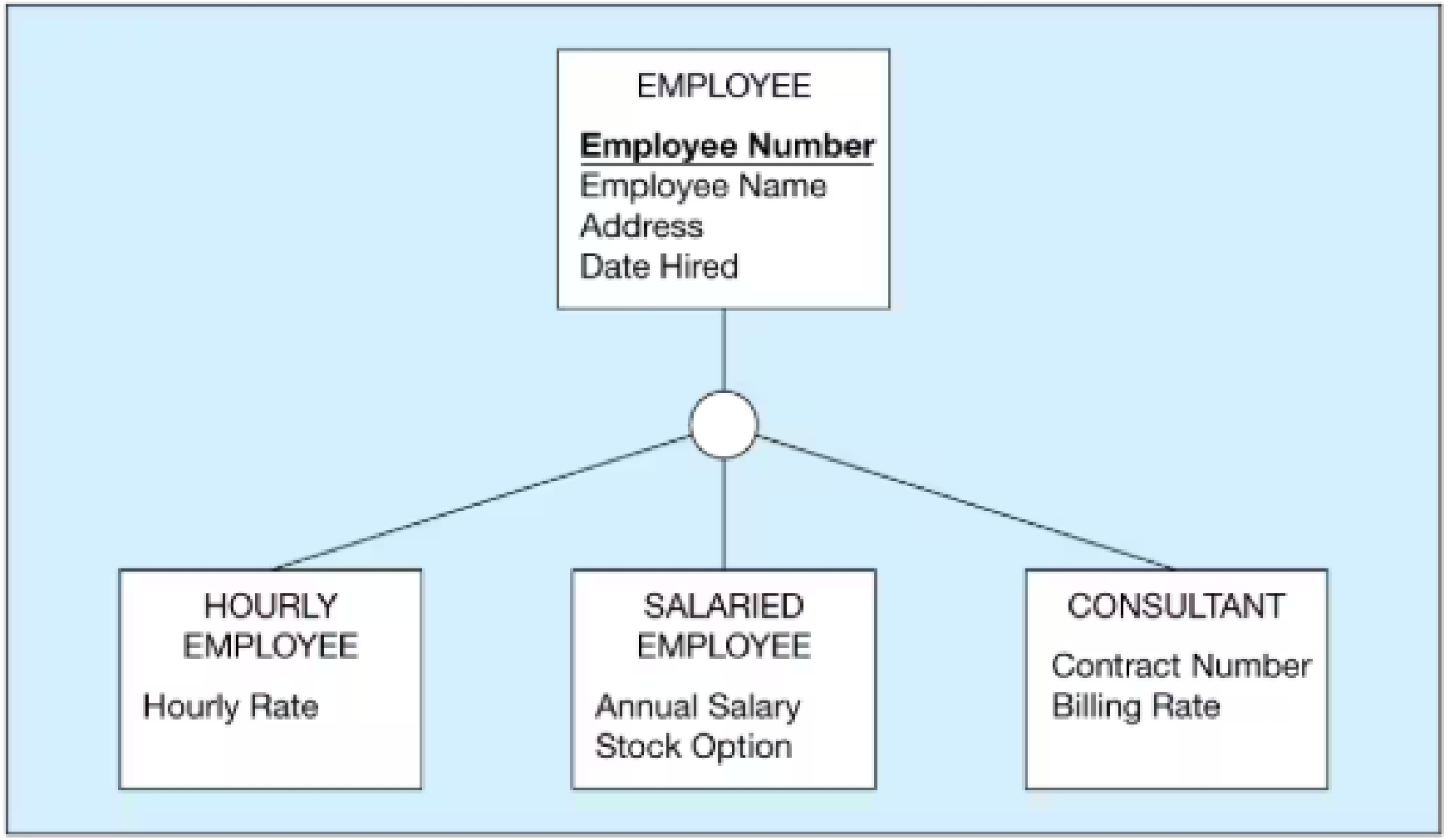
Three kinds of employees, three ways to model them

Shared by all: Employee Number, Employee Name, Address, Date Hired. Then Hourly Rate, or Annual Salary and Stock Option, or Contract Number and Billing Rate.

Choice	How it works	What it costs
One entity, EMPLOYEE	All nine attributes in one box	Every instance carries attributes that do not apply to it. Nulls everywhere, and complicated programs to handle the variations.
Three separate entities	HOURLY EMPLOYEE, SALARIED EMPLOYEE, and CONSULTANT, each complete on its own	The four shared attributes are defined three times. Users must pick the right entity every time.
A supertype with three subtypes	EMPLOYEE holds the shared four. Each subtype holds only its own.	Nothing. Common properties once, distinct properties kept. This is the EER answer.

The employee supertype

Figure 3-2. Nothing is written twice. Employee Number stays with EMPLOYEE, Billing Rate with CONSULTANT.



Attribute inheritance

- A subtype instance is the same entity instance as its supertype instance
- Therese Jones is a CONSULTANT, therefore she is an EMPLOYEE, with a value for every EMPLOYEE attribute including the identifier
- Subtypes inherit all attributes and all relationships of the supertype
- So a subtype never repeats a supertype attribute. Employee Name appears once, at the top.
- Inheritance runs one way. Every subtype member is in the supertype. Whether every supertype member is in some subtype depends on the business. Notation for that is coming.

One person, two rectangles. The second rectangle adds facts. It never repeats them.

When to use subtypes: two triggers

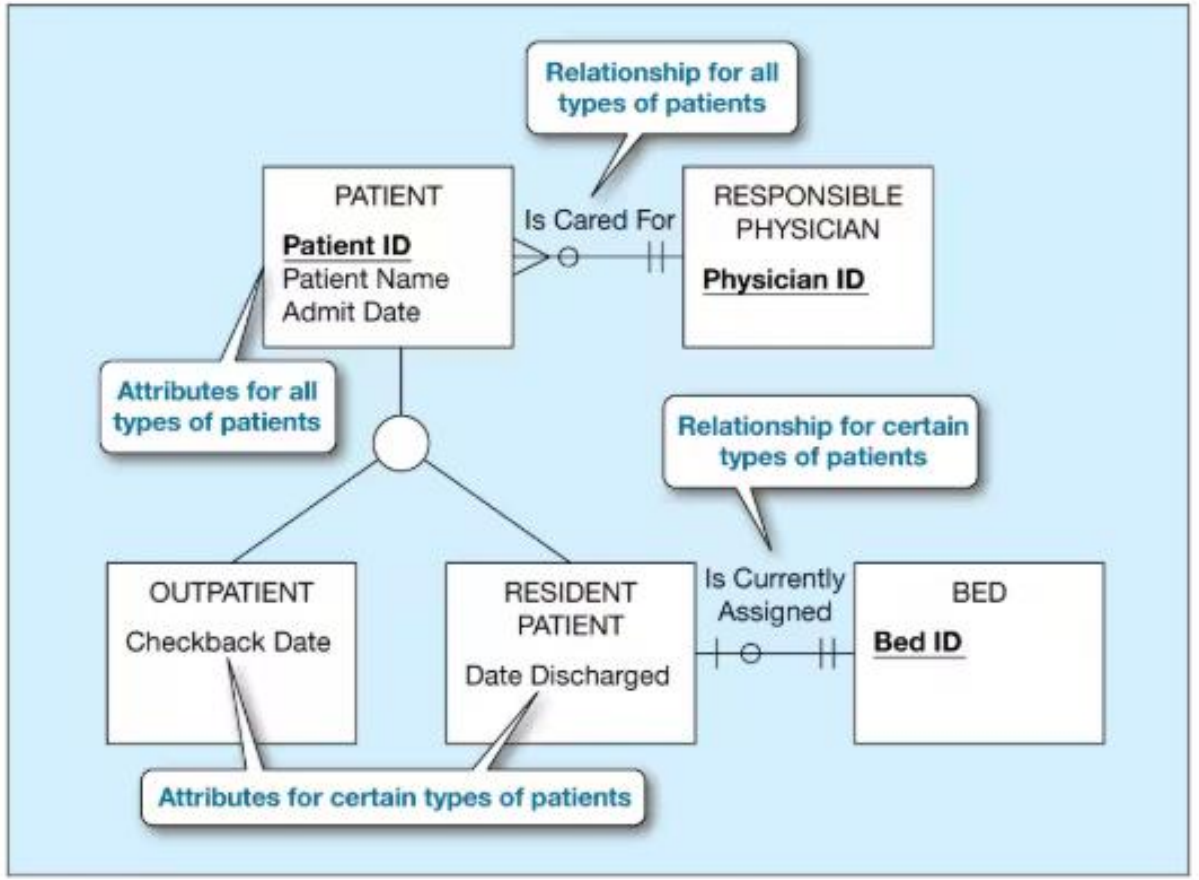
- Trigger 1. Some attributes apply to some, but not all, instances. Hourly Rate.
- Trigger 2. The instances of a subtype take part in a relationship unique to that subtype. Only resident patients get a bed.
- Either trigger is enough. Both together is the textbook case.
- No trigger, no subtype. A kind with nothing of its own stays in the supertype. Remember this when we reach the motorcycle.

A bank. ACCOUNT holds Account Number, Owner, and Open Date. CHECKING ACCOUNT adds Overdraft Limit and the cards issued on it. SAVINGS ACCOUNT adds Interest Rate. Both triggers fire.

Both triggers at once: the hospital

Figure 3-3. Patient ID, Patient Name, and Admit Date belong to everyone.

- Checkback Date belongs to outpatients. Date Discharged belongs to resident patients.
- Only resident patients are assigned a bed, so Is Currently Assigned hangs off the subtype
- Hung there, it can be mandatory. Hung on PATIENT it would have to be optional.
- A bed may or may not hold a patient right now
- Relationship inheritance. Every patient, of either kind, Is Cared For by a responsible physician. Drawn once, at the top.



PART TWO

Finding subtypes and stating the rules

Two ways to find subtypes

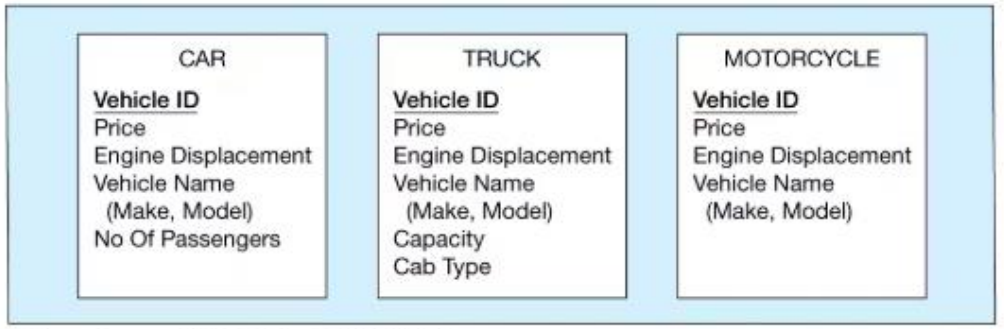
Process	Direction	You start with	You end with	Figure
Generalization	Bottom-up	Several specialized entity types that turn out to overlap	One general supertype plus the specialized subtypes	3-4
Specialization	Top-down	One entity type whose attributes do not all apply to every instance	The same entity as a supertype plus new subtypes	3-5

- Use both. Switch back and forth as the interviews and the old spreadsheets dictate.
- An interview with the warehouse manager (top-down) reveals that some parts are purchased
- Three spreadsheets named Cars, Trucks, and Motorcycles with the same first four columns (bottom-up) reveal one Vehicles list in three hats

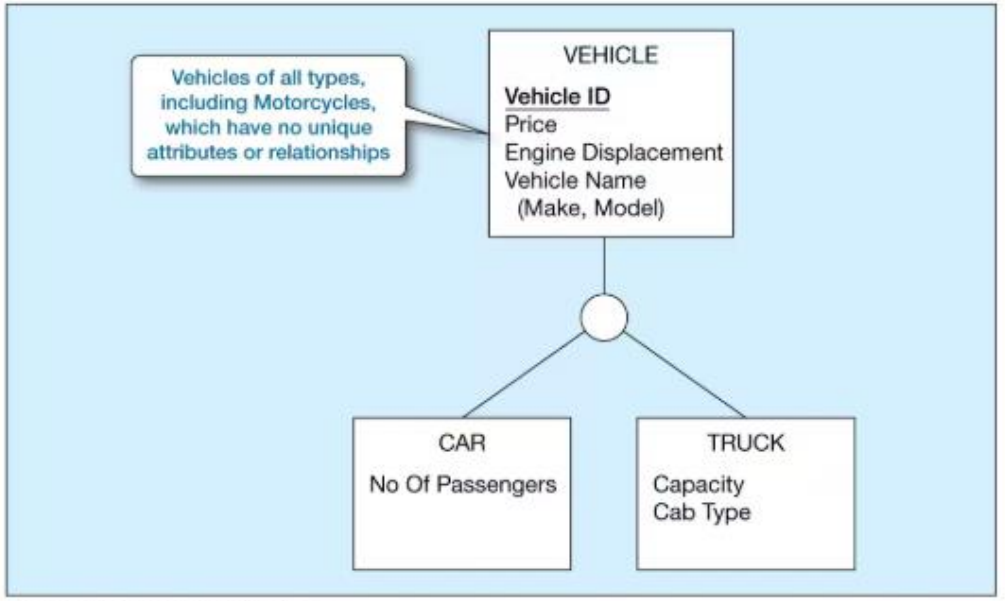
Generalization: bottom-up

Figure 3-4. Three entities that turn out to be one thing wearing three hats.

- CAR, TRUCK, and MOTORCYCLE share Vehicle ID, Vehicle Name (Make, Model), Price, and Engine Displacement
- A shared identifier is the tell. These are versions of one general thing.
- VEHICLE takes the shared four. CAR keeps No Of Passengers. TRUCK keeps Capacity and Cab Type.
- Where is MOTORCYCLE? It has nothing of its own, so it gets no subtype. A motorcycle lives in VEHICLE and nowhere else.
- That means some vehicles are in no subtype. Hold that thought.



(a) Three entity types: CAR, TRUCK, and MOTORCYCLE

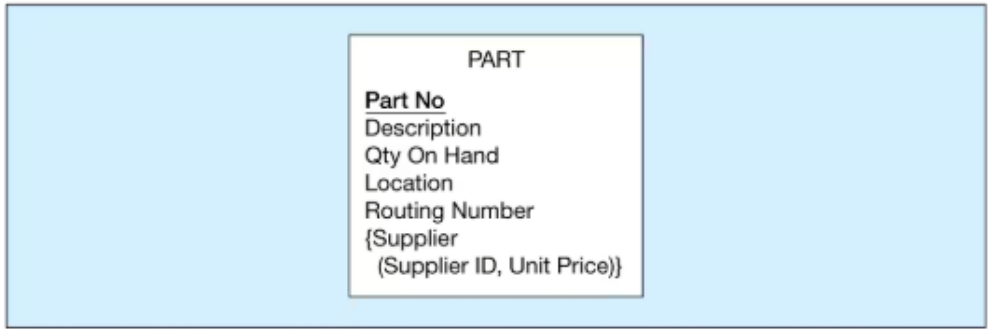


(b) Generalization to VEHICLE supertype

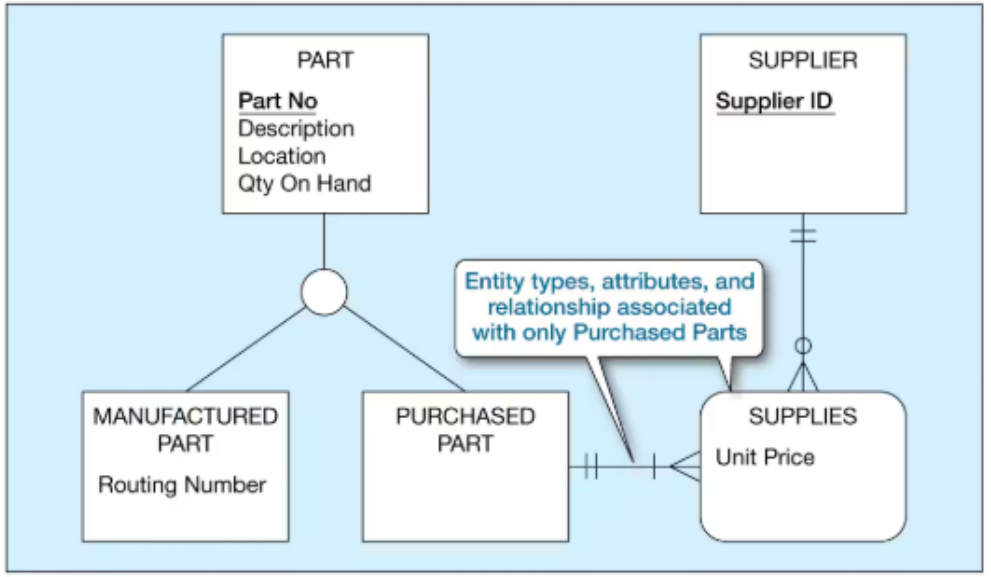
Specialization: top-down

Figure 3-5. Start with PART and its attributes, then split by what the users tell you.

- Parts are manufactured, purchased, or both
- Routing Number applies only to manufactured parts. Supplier ID and Unit Price only to purchased parts. Trigger 1.
- MANUFACTURED PART keeps Routing Number
- The modeler went further: a SUPPLIER entity and a SUPPLIES associative entity, with Unit Price on SUPPLIES, because the price varies by supplier
- Chapter 2 and Chapter 3 working together



(a) Entity type PART



(b) Specialization to MANUFACTURED PART and PURCHASED PART

Two constraints, two questions

Elmasri and Navathe (2021), as cited in the text. Each answer is a business rule.

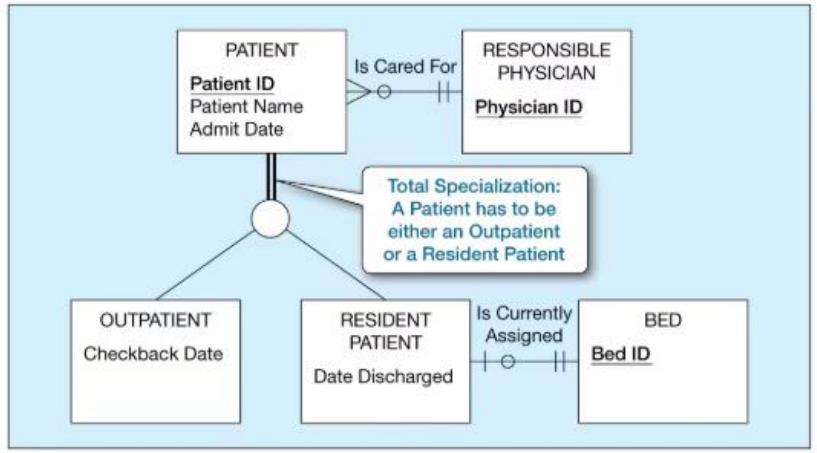
Constraint	The question	If yes	If no
Completeness	Must every supertype instance belong to at least one subtype?	Total specialization: a double line from the supertype to the circle	Partial specialization: a single line
Disjointness	Can one instance belong to two or more subtypes at the same time?	Overlap rule: the letter o in the circle	Disjoint rule: the letter d in the circle

- The two are independent. Any supertype gets one word from each pair.
- Ask both questions every time you draw a circle

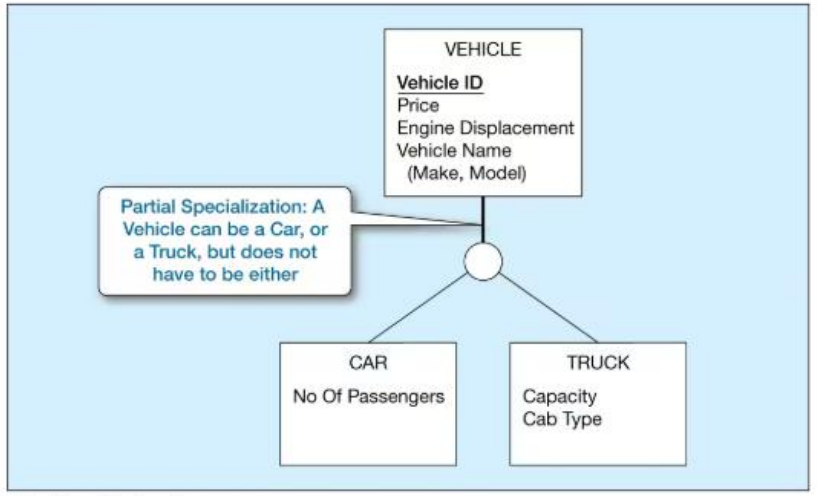
Completeness: total or partial

Figure 3-6. Double line means no leftovers. Single line means leftovers allowed.

- (a) PATIENT. Every patient is an outpatient or a resident patient. This hospital has no third kind.
- Insert a patient and a subtype instance goes in with it. Insert a resident and a bed assignment goes in too.
- (b) VEHICLE. Cars go in CAR, trucks in TRUCK, and motorcycles stay in VEHICLE alone.
- Visio calls total specialization “Category is complete”
- The circle in (b) is empty on purpose. Disjointness is the next figure.



(a) Total specialization rule

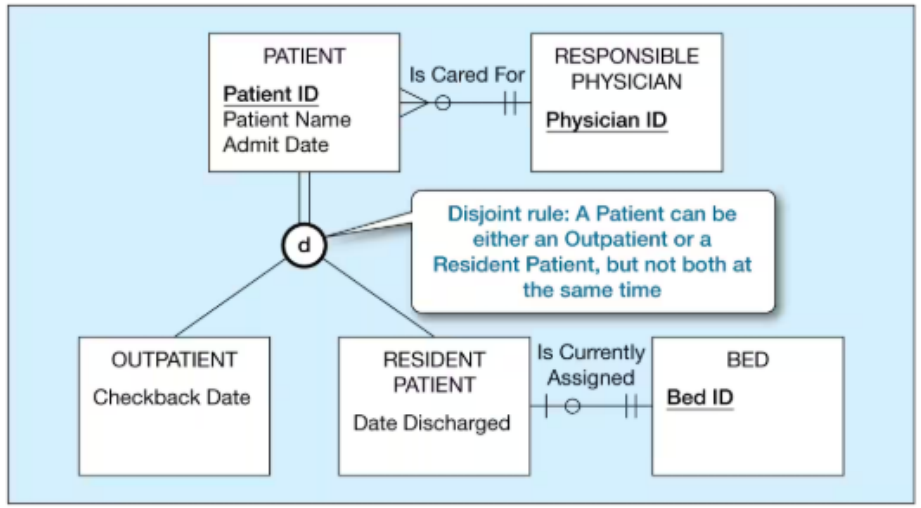


(b) Partial specialization rule

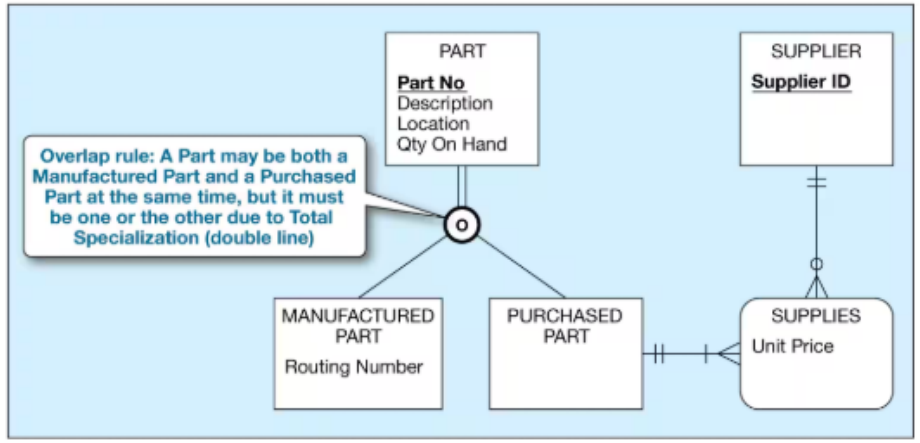
Disjointness: disjoint or overlap

Figure 3-7. The letter in the circle.

- (a) d. At any given time a patient is one type, never both. The type can change next month.
- (b) o. A part can be manufactured and purchased at once.
- An instance of PART is a part number, not a physical piece. Part 4000: 250 on hand, 100 made, 150 bought.
- (b) also carries the double line. Every part is one or the other, and may be both.
- Visio has no symbol for this. Type the letter into the circle.



(a) Disjoint rule



Every combination, with the chapter's examples

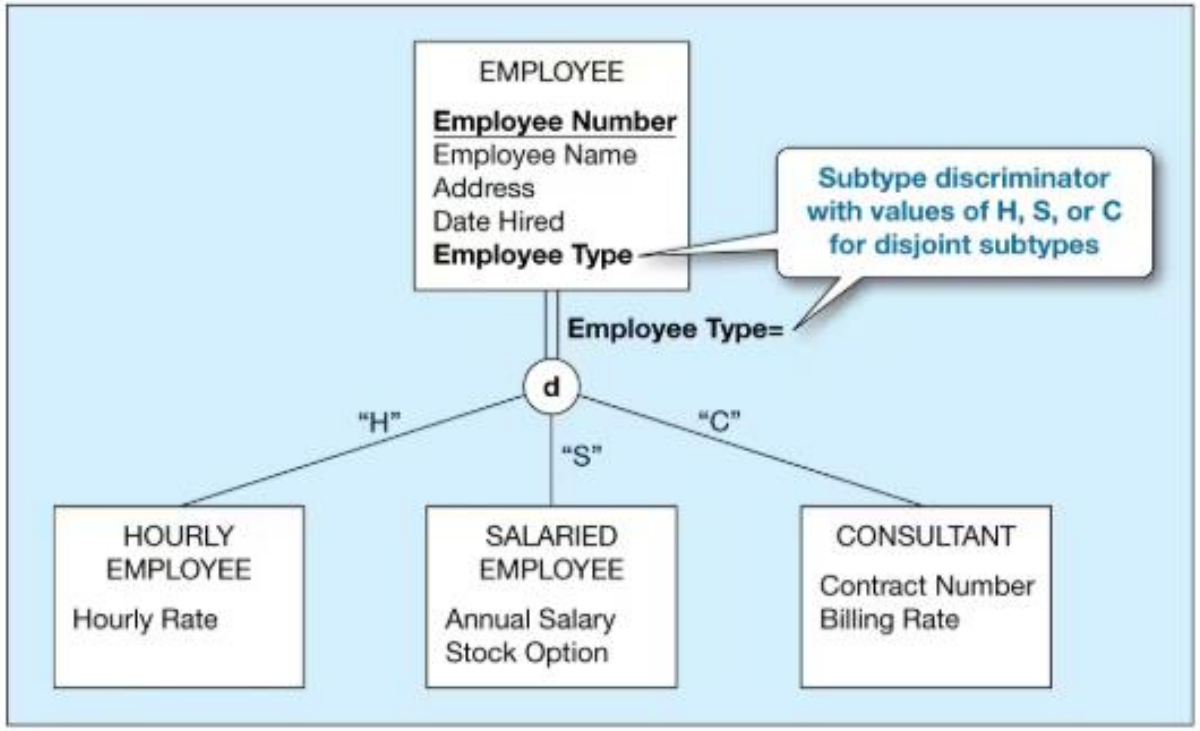
	Disjoint (d): never both at once	Overlap (o): can be both at once
Total (double line): every instance is in a subtype	PATIENT: outpatient, resident (3-7a) EMPLOYEE: hourly, salaried, consultant (3-8) STUDENT: graduate, undergrad (3-10)	PART: manufactured, purchased (3-7b) PERSON: employee, student, alumnus (3-10) CUSTOMER: regular, national account (3-12)
Partial (single line): some instances are in no subtype	EMPLOYEE: faculty, staff (3-10) EMPLOYEE: management, union (3-12)	Not drawn in the chapter. A library PERSON: borrower, volunteer, both, or neither.

- Only one subtype, as VENDOR to SUPPLIER? No letter. There is nothing to overlap with.
- Disjoint means never both at once, not never changes. A union employee promoted into management switches boxes.

Subtype discriminator: disjoint subtypes

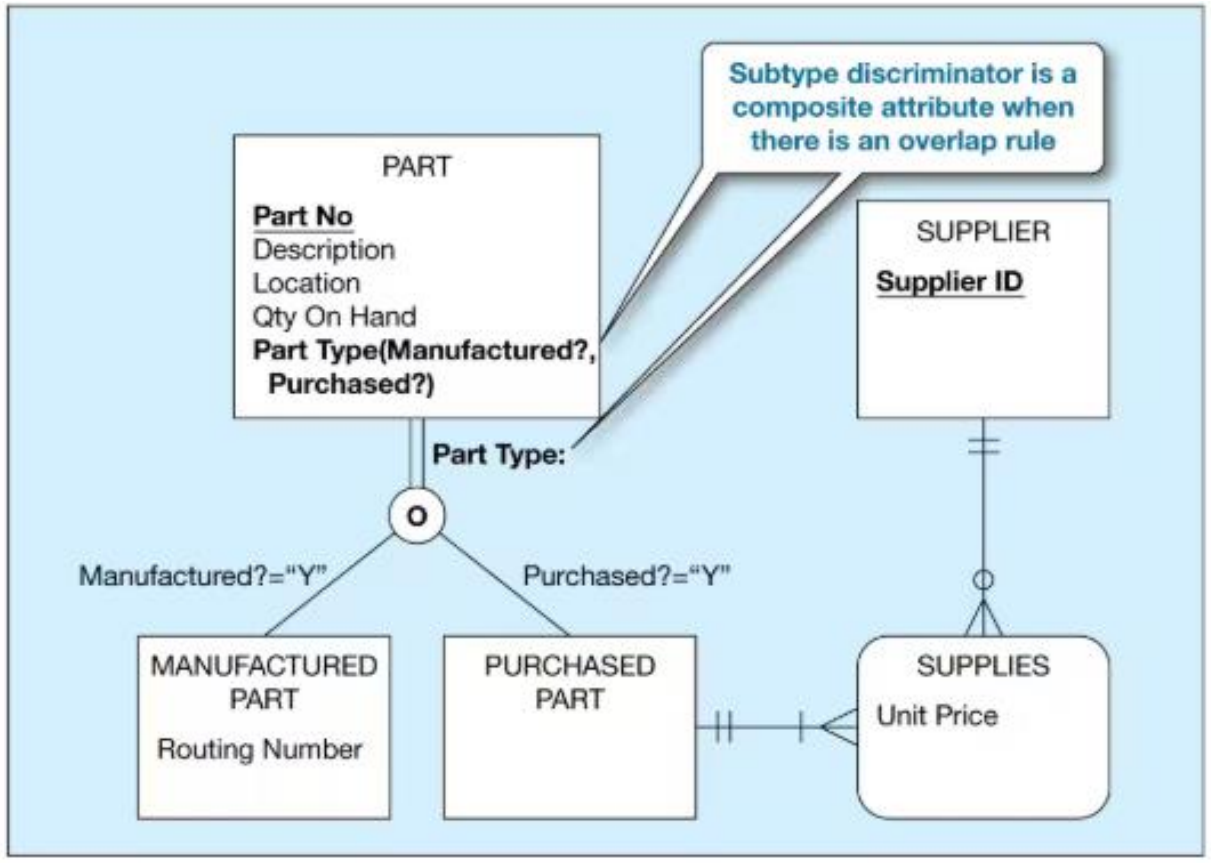
Figure 3-8. Employee Type = H, S, or C.

- Which subtype does a new supertype instance go into? An attribute of the supertype decides.
- Employee Type= sits on the line to the circle. The value that picks each subtype sits on that subtype's line.
- Employee Type="S" sends the instance into SALARIED EMPLOYEE
- The discriminator lives in the supertype, never in a subtype
- One code, one answer. That works only because the subtypes are disjoint.



Subtype discriminator: overlapping subtypes

Figure 3-9. One yes/no component per subtype.



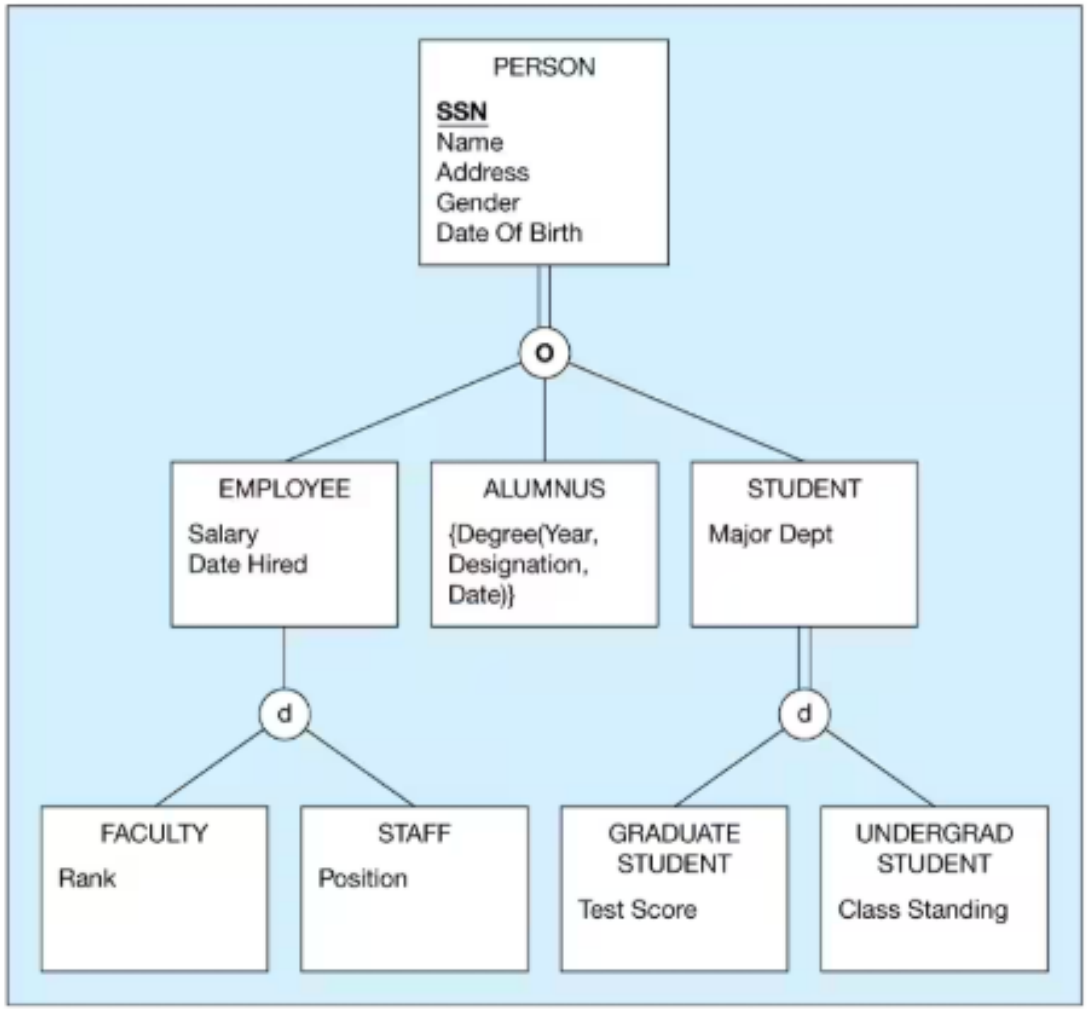
Type of Part	Manufactured?	Purchased?
Manufactured only	"Y"	"N"
Purchased only	"N"	"Y"
Purchased and manufactured	"Y"	"Y"

- One code cannot send an instance into two subtypes
- Part Type is composite: Manufactured? and Purchased?, each Y or N
- Manufactured?="Y" fills MANUFACTURED PART. Purchased?="Y" fills PURCHASED PART. Both Y, both subtypes.
- Works for any number of overlapping subtypes

A hierarchy: the university

Figure 3-10. Each subtype has exactly one supertype. Discriminators omitted for clarity.

- Root: PERSON with SSN, Name, Address, Gender, Date Of Birth
- First level: EMPLOYEE, STUDENT, ALUMNUS. Total and overlap. Nobody else matters to the university, and one person can be all three.
- EMPLOYEE splits into FACULTY and STAFF: partial (student assistants exist) and disjoint
- STUDENT splits into GRADUATE and UNDERGRAD: total and disjoint
- Say the STUDENT rules out loud. Every student is one kind. No student is both.



Two rules about attributes in a hierarchy

- Put each attribute at the highest level where it applies. SSN on the root, Date Hired on EMPLOYEE, Rank on FACULTY. That lets the most subtypes share it.
- Inheritance runs all the way up the chain, not just one step

What one FACULTY instance carries

Attributes	Inherited from	Level
SSN, Name, Address, Gender, Date Of Birth	PERSON	Root
Date Hired, Salary	EMPLOYEE	First level
Rank	FACULTY (its own)	Second level

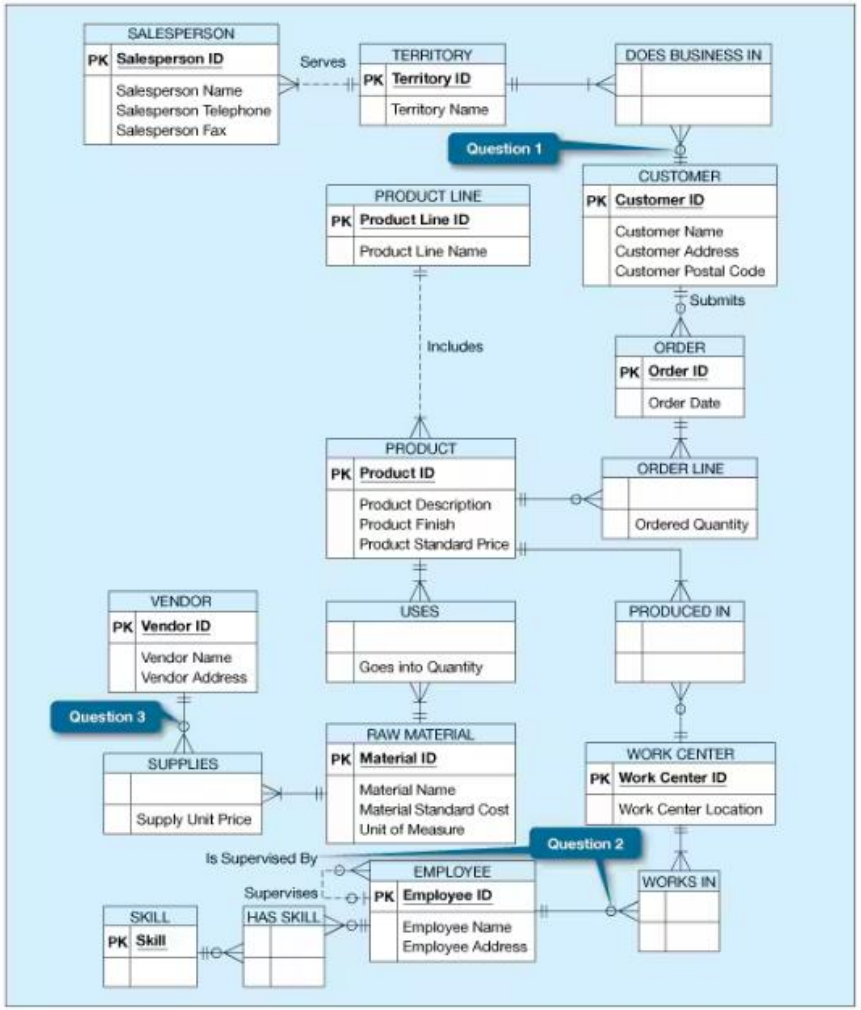
Wichita State keeps you as a PERSON who is a STUDENT. Work on campus or graduate, and EMPLOYEE or ALUMNUS facts get added. Your name is never stored twice.

PART THREE

Pine Valley Furniture and entity clusters

Pine Valley, as we left it

Figure 3-11. Three cardinalities that raise questions.



- Question 1. Why do some customers do business in no sales territory?
- Question 2. Why do some employees supervise nobody, why are they not all supervised, and why do some work in no work center?
- Question 3. Why do some vendors supply no raw materials?
- The optional marks are correct. The reasons are invisible. That is what subtypes fix.

What the investigation found

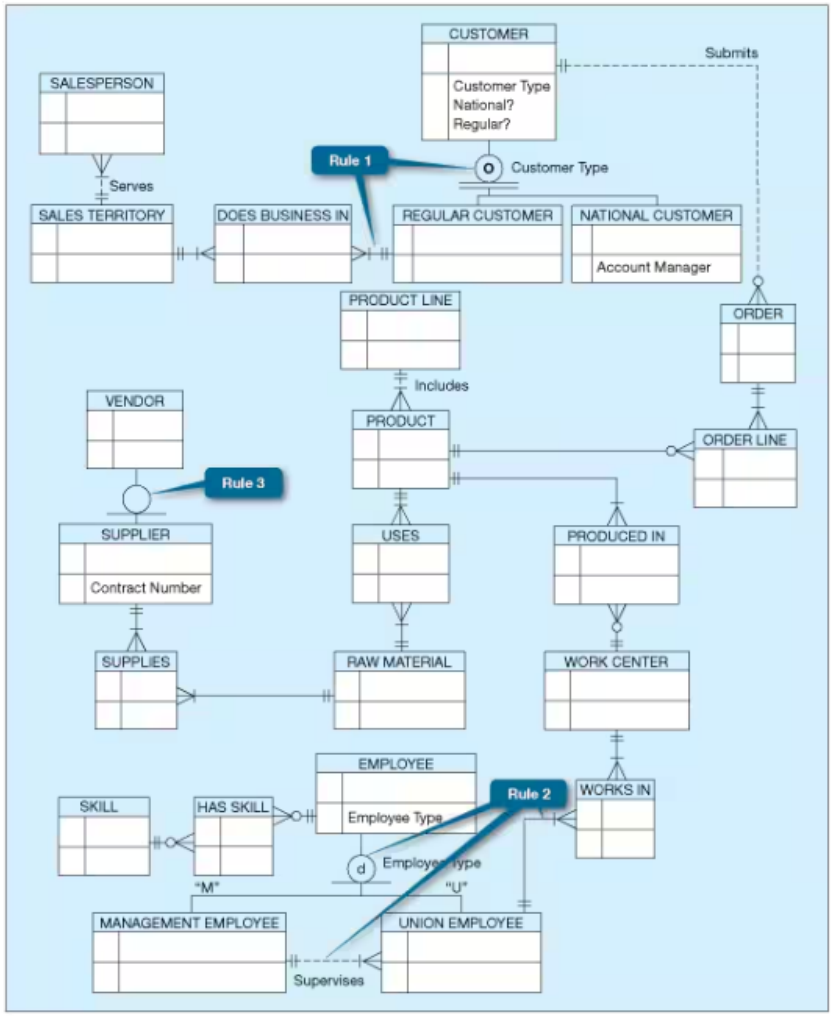
- Rule 1. Customers. Two types, regular and national account. Only regular customers do business in territories, and a territory needs at least one. A national account has an account manager. A customer can be both.
- Rule 2. Employees. Management and union are special types. Only union employees work in work centers. Management supervises union. Other kinds of employees exist. Promotion moves a union employee into management.
- Rule 3. Vendors. Not every vendor has supplied anything. A vendor gets a contract number once it becomes an official supplier.

Each rule answers one of the three questions. Now translate each one into notation.

Pine Valley, enhanced

Figure 3-12. Most attributes are left off so the changes stand out.

- CUSTOMER: total, overlap, Customer Type (National?, Regular?). Does Business In moves down to REGULAR CUSTOMER.
- EMPLOYEE: partial, disjoint, Employee Type M or U. Works In moves to UNION EMPLOYEE and becomes mandatory. Supervises now runs management to union.
- VENDOR: partial, one subtype, so no letter. Contract Number and Supplies move to SUPPLIER. The minimum next to RAW MATERIAL becomes one.
- The figure says NATIONAL CUSTOMER. The text says NATIONAL ACCOUNT CUSTOMER. Same box.



What changes when E-R becomes EER

Kind of change	At Pine Valley
New entities: supertypes and subtypes	REGULAR CUSTOMER, NATIONAL ACCOUNT CUSTOMER, MANAGEMENT EMPLOYEE, UNION EMPLOYEE, SUPPLIER
New attributes, including discriminators	Customer Type, Employee Type, Account Manager, Contract Number
Minimum cardinalities move from optional to mandatory	Works In from UNION EMPLOYEE, Supplies toward RAW MATERIAL
Relationships move from the supertype to the subtype that takes part	Does Business In, Works In, Supervises, Supplies

Given one rule, name the change it causes. “Only union employees work in work centers” moves Works In down and makes it mandatory.

A data model is not a database design

- The EER diagram is a conceptual picture of the data the business needs
- It does not map one-for-one onto tables. A designer may still put every customer in one table.
- Our job now is to state every rule that governs the data. Storage and speed come later.
- Chapter 4 is where the tables get decided

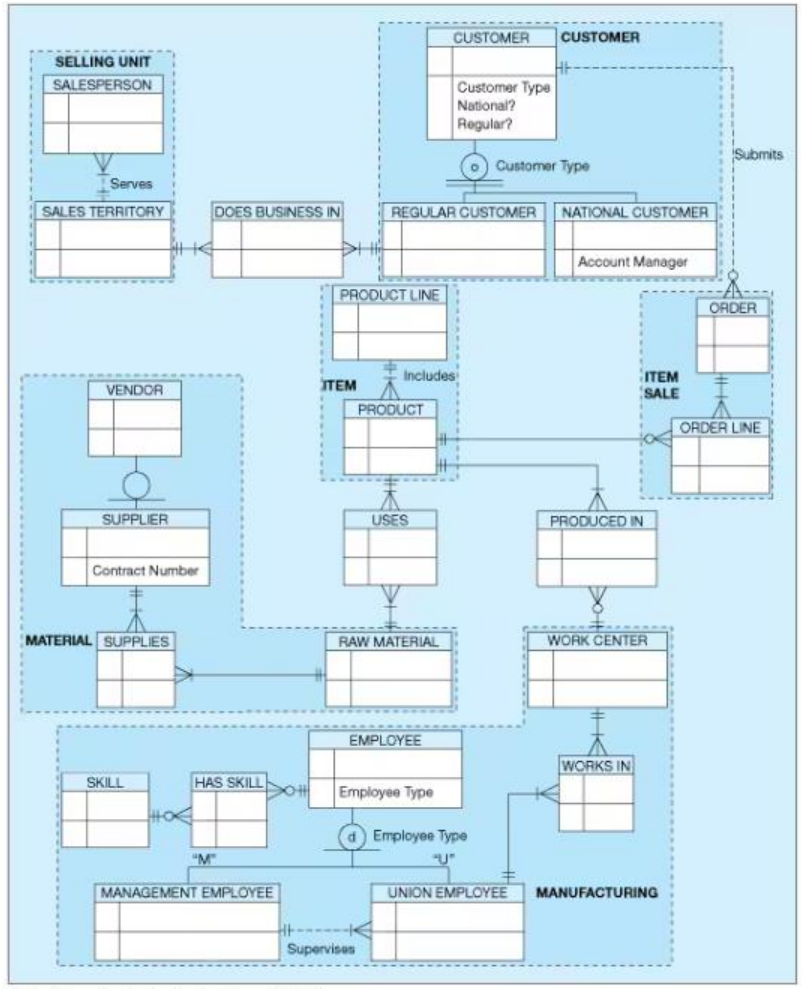
The drawing says kitchen and pantry. The builder may pour one slab under both.

Entity clustering: the problem and the idea

- Some enterprise models hold more than 1,000 entity types and relationships
- A wall-sized printout helps nobody. Almost nobody needs the whole picture.
- Systems analysis solves this with functional decomposition (Valacich and George, 2020, as cited in the text)
- Data modeling borrows it: several linked diagrams, each showing one segment for one audience. Segments may overlap.
- An entity cluster is a set of entity types and their relationships grouped into one abstract entity type (Teorey, 1999, as cited in the text)
- Clusters can be clustered again. Zoom out, zoom in, same territory.

Marking the clusters

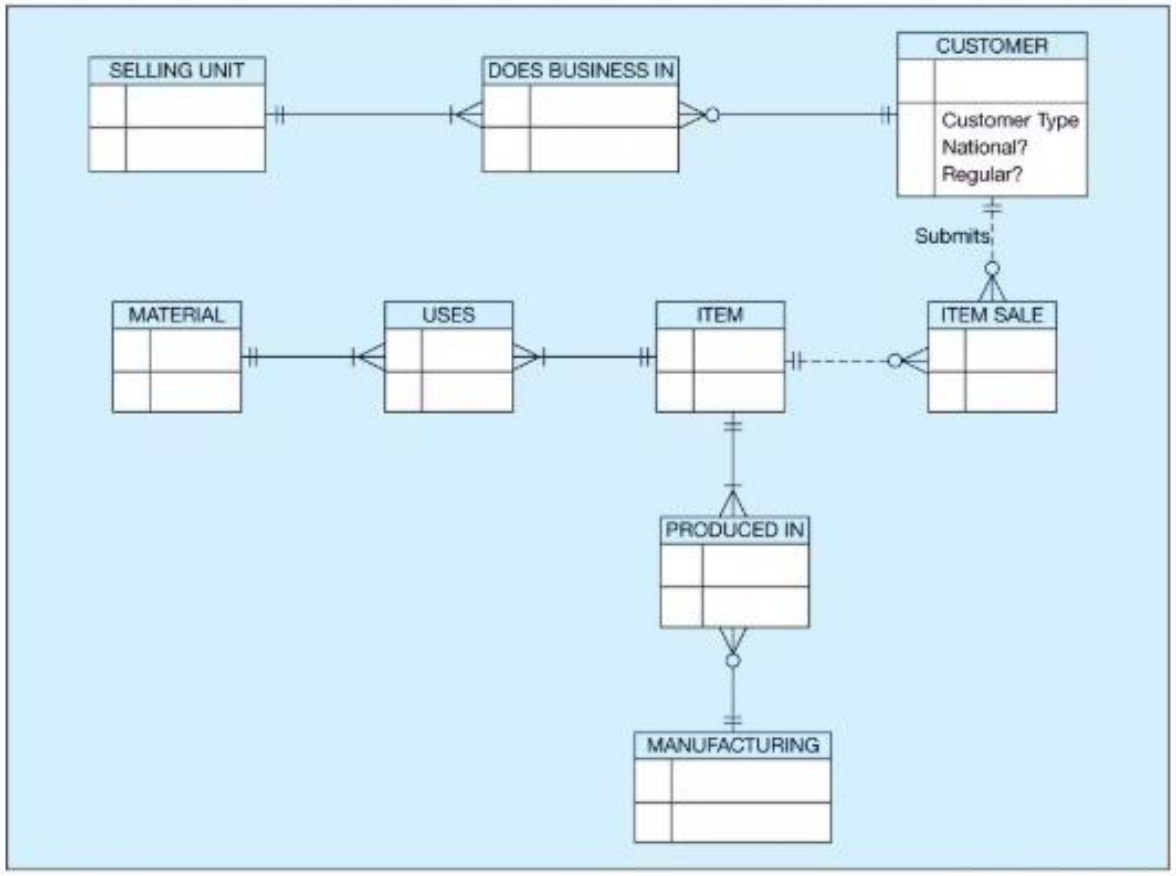
Figure 3-13(a). The full Pine Valley EER diagram with shaded areas around possible clusters.



(a) Possible entity clusters (using Microsoft Visio)

The cluster diagram

Figure 3-13(b). Six clusters and the relationships between them. Nothing else.

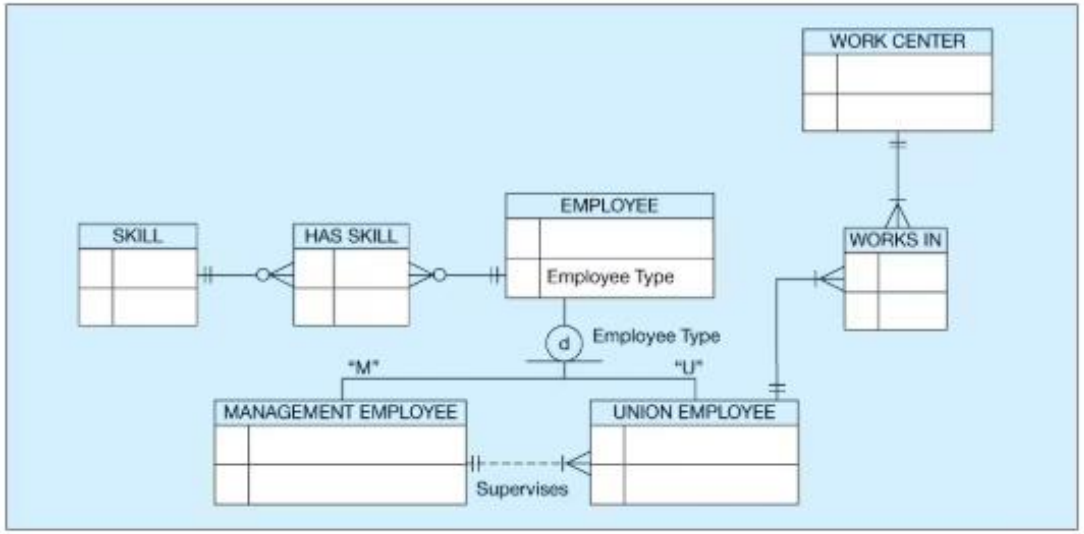


(b) EER diagram for entity clusters (using Microsoft Visio)

- SELLING UNIT: SALESPERSON, SALES TERRITORY, Serves
- CUSTOMER: the supertype, its subtypes, and their connection
- ITEM SALE: ORDER, ORDER LINE, and the relationship between them
- ITEM: PRODUCT LINE, PRODUCT, Includes
- MANUFACTURING: WORK CENTER, EMPLOYEE and its subtypes, Works In, Supervises
- MATERIAL: RAW MATERIAL, VENDOR, SUPPLIER, Supplies

Opening one cluster

Figure 3-14. MANUFACTURING, exploded for the people who run the shop floor.



- An inventory manager reads Figure 3-13(b) to see how manufacturing ties to items, then this figure for the detail. The selling structure never appears.
- Three ways to form a cluster: abstract a supertype and its subtypes (CUSTOMER), combine directly related entities (the other five), or combine a strong entity with its weak entities
- No single right clustering. Group for the audience. Move ORDER into CUSTOMER and ORDER LINE into ITEM, and ITEM SALE disappears.

PART FOUR

Packaged data models and the models behind analytics

Why organizations buy a data model

- Silverston (1998), as cited in the text: the modeler as artisan gives way to the modeler as engineer
- Universal or industry-specific, sold as “logical data models.” They are EER diagrams.
- The bet: enterprises in one industry share their underlying data structures
- Overstock.com, studied by the authors: the purchased model was about 80 percent right before customization and cost about one modeler-year
- Proven components, fewer misses, a ready-made checklist for requirements, easier to evolve, easier to read, easier to share data across an industry
- A universal data model is a starting point, never the “right” model

A house plan from a catalog instead of an architect starting from a blank page. You still rename the study and drop the third garage.

How the project changes

- You buy metadata: every entity, attribute, and relationship, defined and typed, in a tool such as erwin. The tool can generate the SQL.
- Find the parts that apply. Entities, then attributes, then relationships. Model the future, not just today.
- Rename the generic terms to the organization's own words
- Map the package to the current databases. Some elements will not map in either direction, and that is expected.
- The package's rules are deliberately loose. “At most one customer per order” is your rule to add.
- Users react to a concrete diagram, not a wish list. Build in phases, and use entity clusters to show the phases.
- It takes more skill, not less

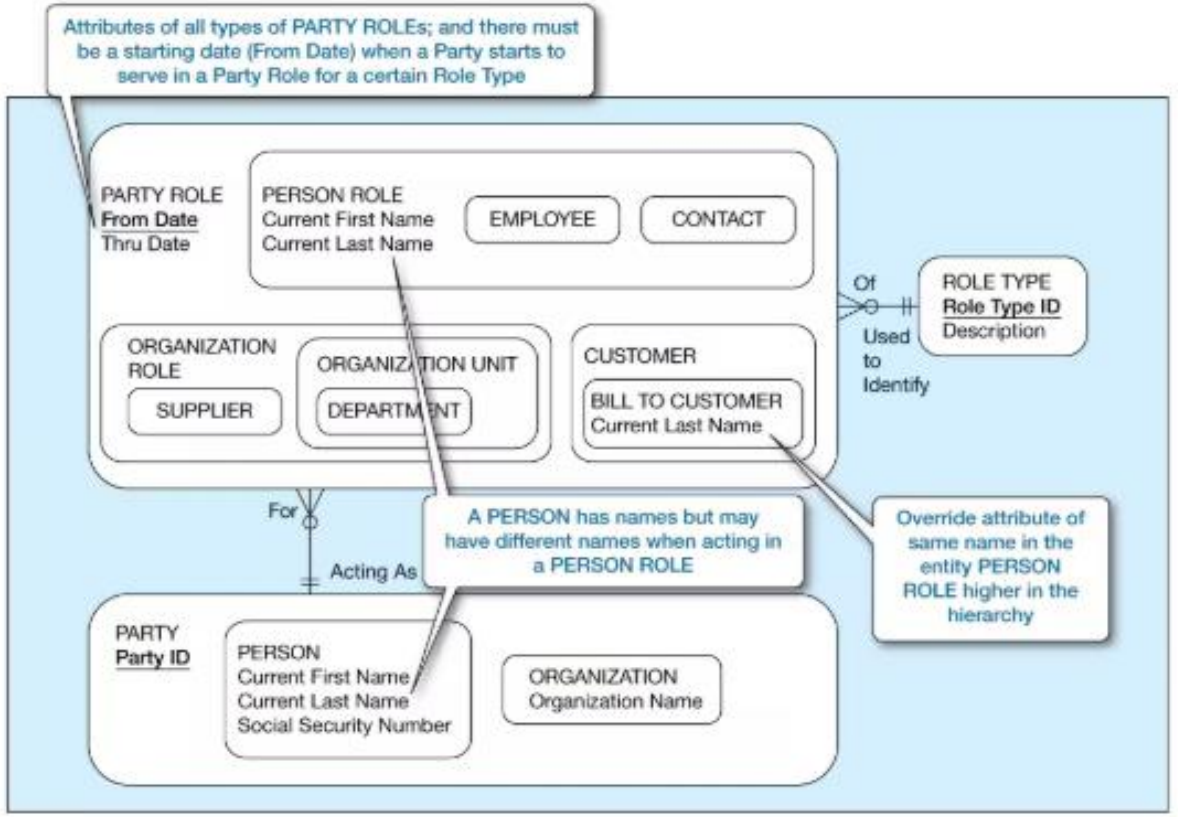
What the old databases will teach you

Issue	What it means	What to do
Overloaded columns	One attribute used for several purposes over the years, usually undocumented	Expect surprises during migration. Some data are dead, and some hide requirements nobody wrote down.
Empty attributes	Missing values from bugs, entry errors, or products older than the attribute	Ask why. Sparse data hint at optional attributes, or at subtypes.
Hidden structure	Special-project shortcuts never undone, rule changes nobody dated	Data profiling: statistics that expose outliers and shifts in the data over time
Unverified rules	The package guesses most rules right, but every one must be confirmed	Walk each element with a subject matter expert. The hardest part of the job.

PARTY, PARTY ROLE, and ROLE TYPE

Figure 3-15(a). The core of every universal model, in the inside-the-box notation.

- PARTY: the actor, a PERSON or an ORGANIZATION
- PARTY ROLE: that actor playing a role from one date to another. EMPLOYEE, CONTACT, SUPPLIER, DEPARTMENT, CUSTOMER.
- ROLE TYPE: the description of a kind of role, written once for everyone who plays it
- A role can override the party's values. A different Current Last Name while acting as BILL TO CUSTOMER.
- The person is the party. Each hat is a role, with the dates it was worn.

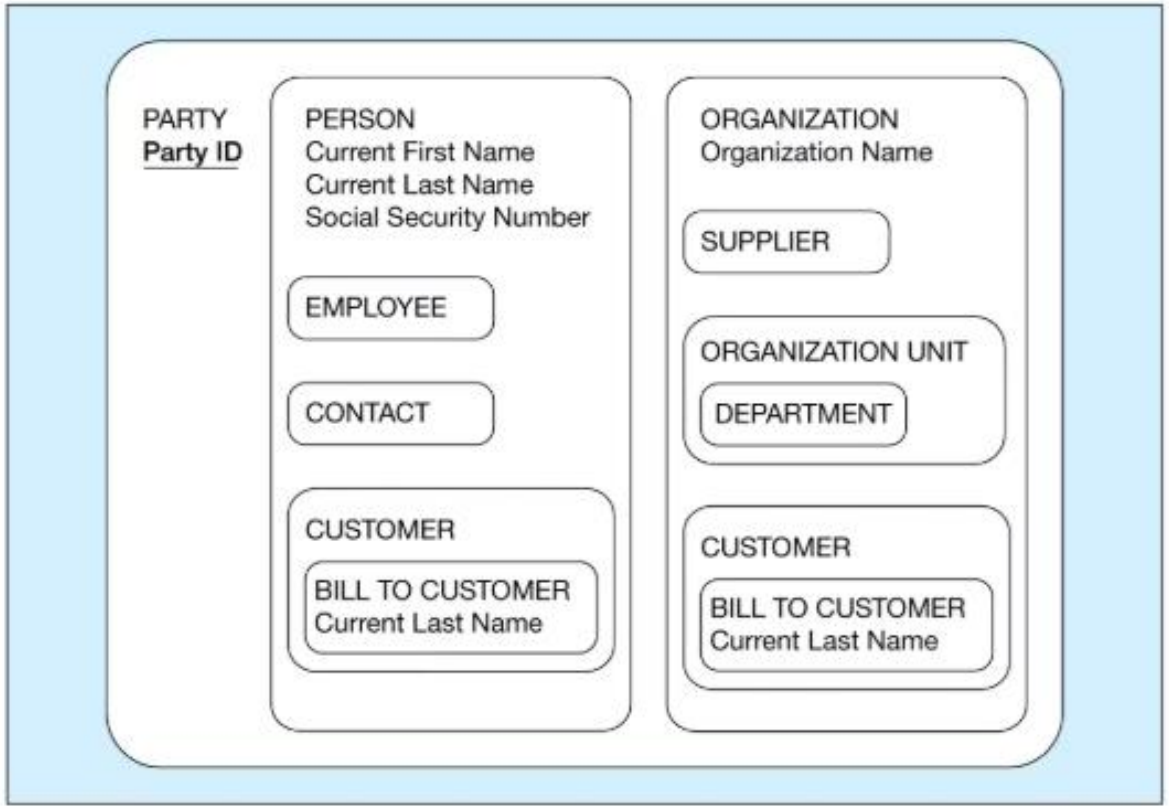


(a) Basic PARTY universal data model

Why not one long list of subtypes?

Figure 3-15(b). The alternative a beginner would draw.

- One PARTY supertype, partial and overlapping, with a subtype for every role
- On paper it carries the same meaning
- But it sews every hat onto the person. Actor and role collapse into one box.
- Keeping them apart makes the model more general, so it fits more organizations. That is the whole point of a universal model.

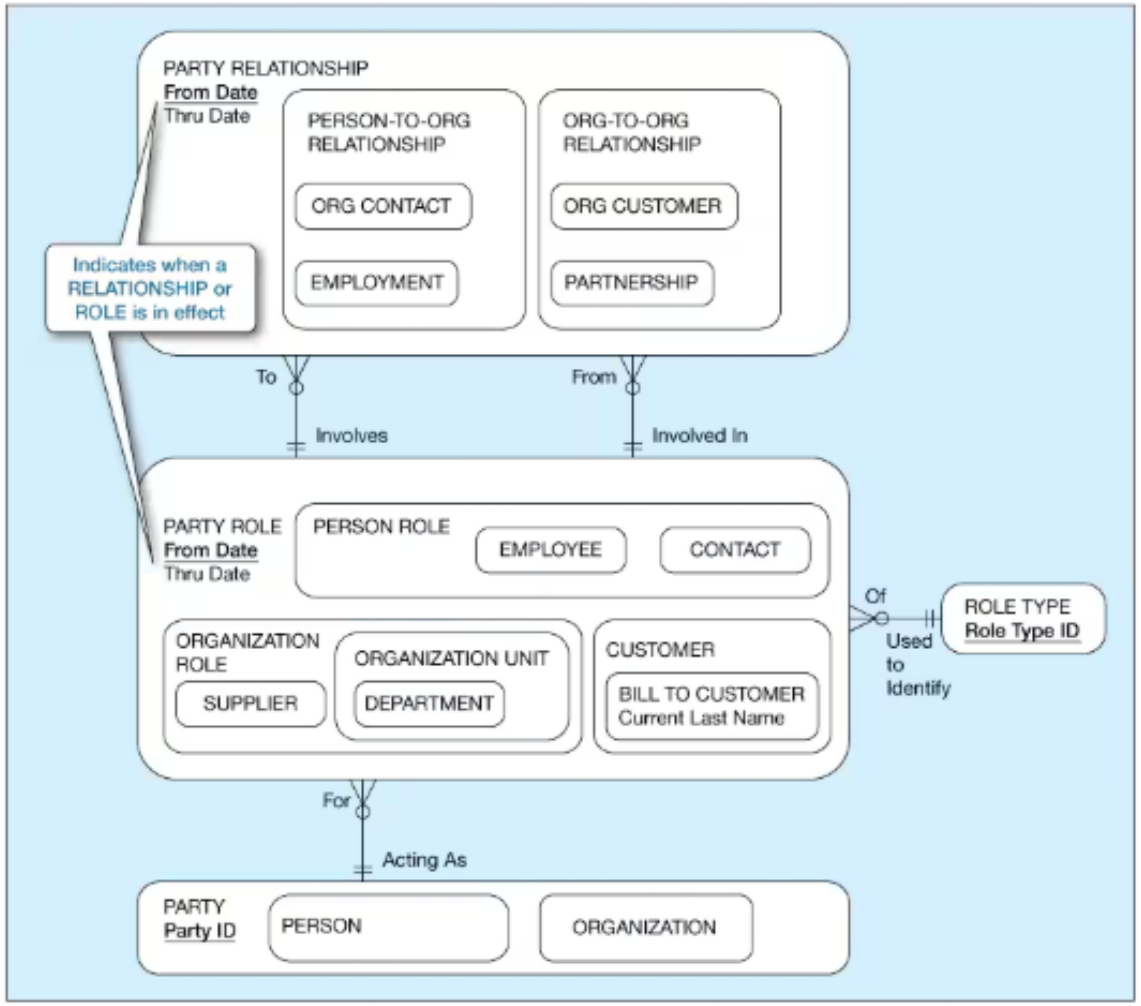


(b) PARTY supertype/subtype hierarchy

Relationships between parties

Figure 3-16. PARTY RELATIONSHIP, an associative entity with From Date and Thru Date.

- Relates any number of parties in the roles they play. Subtypes: EMPLOYMENT, ORG CONTACT, ORG CUSTOMER, PARTNERSHIP.
- EMPLOYMENT links a PERSON in the EMPLOYEE role to an ORGANIZATION in the ORGANIZATION UNIT role, for a period
- It is really a unary many-to-many among PARTY ROLEs. Flexible, and vague.
- Nothing stops EMPLOYMENT from linking two suppliers. Nothing limits an employee to one unit at a time.
- Stricter rules end up in stored procedures or programs, or in a very busy diagram



Four older alternatives to E-R

Model	Core idea	Where it lives
Object-oriented	Objects with their own attributes and methods. Handles graphs and trees.	Software engineering. The book's web Chapter 13.
Semantic	Models meaning the way people understand it. EER is one way to build one.	Knowledge management systems
UML	A general-purpose modeling language. Data drawn as class diagrams.	Object-oriented software development
Hierarchical	A tree of parent-child records	A few old DBMSs. Popular in the 1960s and 1970s.

None of them replaces E-R or EER. None captures the full range of business rules.

Models built for analytics

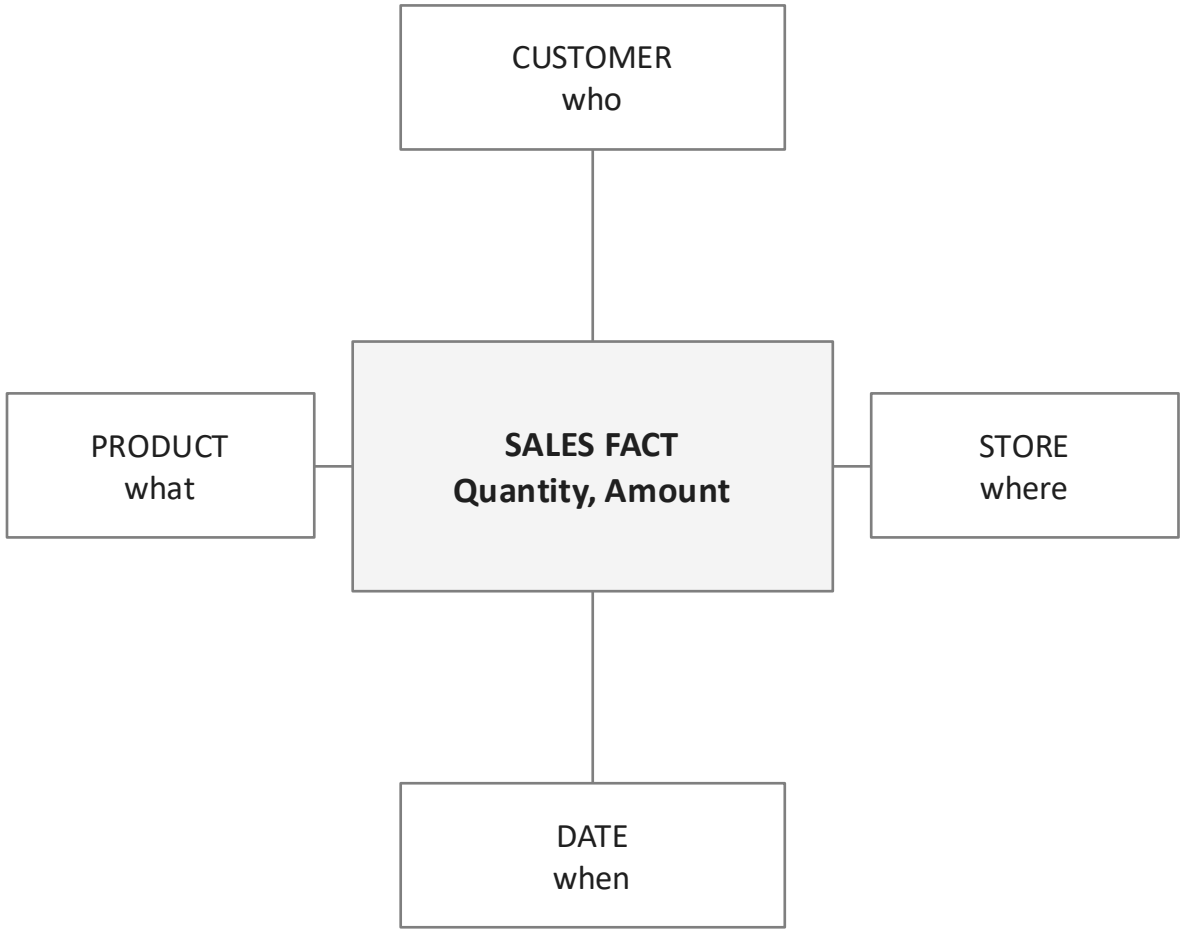
The right one depends on the application, the data, and the performance the system needs.

Model	What it is	Good for	Chapter
Star schema	A fact table of measures ringed by dimension tables	OLAP: aggregate and drill down. Born in data warehousing.	10
Snowflake	A star whose dimensions are normalized into several tables	Less redundancy, harder queries	10
Columnar	Data stored by column, not by row	Scans and aggregates over very large tables	9
NoSQL	Key-value, graph, and document databases	Large volumes of unstructured or semi-structured data, real-time analytics	9
Time series	Time-stamped points at regular intervals: prices, sensors, logs	Real-time monitoring and analysis	none
Graph	Nodes and relationships both hold data	Social networks, fraud detection, recommendations	9
Document	Self-describing JSON or BSON documents, each with its own structure	Flexible, scalable stores with no cross-document rules	9

Chapter 9 is on our schedule. Chapter 10 is not, so this is your look at the star schema. Graph and document are NoSQL families.

The star schema, in one picture

Drawn for this course. The fact sits in the middle, the context around it.



- The fact is a line on a receipt: how many, how much
- The dimensions answer who, what, where, and when
- “Office chair sales in Kansas last quarter” is one fact table filtered by three dimensions
- Snowflake: split PRODUCT into PRODUCT and PRODUCT LINE, and so on, the way Chapter 4 normalizes
- Not a substitute for the EER model. It answers questions fast. It does not state the rules of the business.

Before next class

- Tonight's in-class exercise. Turn it in for participation credit.
- Study guide. The Chapter 3 guide on Blackboard walks every figure from tonight, with self-check questions.
- Blackboard tasks. Chapter 3 quiz, matching exercise, and the written assignment. Graduate students add the extension.
- Draw.io habit. A circle with d or o inside, a double line for total specialization, shared attributes in the supertype only.
- Exam 1 is September 23. Chapters 1 through 4.

Next: Chapter 4. Supertypes and subtypes become tables, and normalization begins.